

习题课 第三章,

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5. 解: $X \sim \text{Exp}(1)$, $\therefore E(X) = 1$

$$E(2X) = 2E(X) = 2 \times 1 = 2.$$

$$E(e^{-2X})$$

$$\left(\underline{E(g(X))} = \int_{-\infty}^{+\infty} g(x) f(x) dx \right).$$

$$E(e^{-2X}) = \int_{-\infty}^{+\infty} e^{-2x} \cdot f(x) dx$$

$$= \int_0^{+\infty} e^{-2x} \cdot e^{-x} dx$$

$$= \frac{1}{3} \int_0^{+\infty} 3e^{-3x} dx = 1/3.$$

7. 解: $X \sim \text{Exp}(\frac{1}{2}), Y \sim \text{Exp}(\frac{1}{4})$

$$E(X) = \frac{1}{2}, E(Y) = \frac{1}{4}$$

$$E(X+Y) = E(X) + E(Y) = 3/4.$$

$$E(2X - 3Y^2) = 2E(X) - 3E(Y^2).$$

$$D(Y) = E(Y^2) - (EY)^2$$

$$\therefore E(Y^2) = \frac{D(Y) + (E(Y))^2}{1} \\ = \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2 = 1/8.$$

$$\therefore E(2X - 3Y^2)$$

$$= 2E(X) - 3E(Y^2)$$

$$= 1 - 3/8$$

$$= 5/8.$$

9. 解: $E(X) = \int_{-\infty}^{+\infty} x f(x) dx$

$$= \int_{-1}^1 \frac{1}{\pi} \cdot \frac{1}{\sqrt{1-x^2}} \cdot x dx$$

$$= 0.$$

$$D(X) = E(X^2) - (E(X))^2 = E(X^2)$$

而 $E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx$

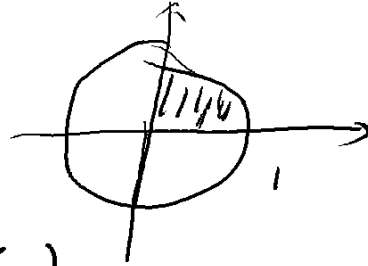
$$= \frac{1}{\pi} \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx. \quad \left(\begin{array}{l} \text{也可令} \\ x = \sin \theta \end{array} \right)$$

$$= \frac{2}{\pi} \int_0^1 \frac{x^2 - 1 + 1}{\sqrt{1-x^2}} dx$$

$$= \frac{2}{\pi} \cdot \left(\int_0^1 \frac{-1}{\sqrt{1-x^2}} dx + \int_0^1 \frac{1}{\sqrt{1-x^2}} dx \right)$$

$$\text{其中: } \int_0^1 \sqrt{1-x^2} dx = \frac{\pi}{4}$$

$$y = \sqrt{1-x^2} \Rightarrow x^2 + y^2 = 1$$



$$\begin{aligned} \therefore E(X^2) &= \frac{2}{\pi} \left(-\frac{\pi}{4} + \frac{\pi}{2} \right) \\ &= \frac{1}{2} \end{aligned}$$

$$\therefore D(X) = E(X^2) = \frac{1}{2}.$$

10. 解:

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$= \int_0^{+\infty} x \cdot \frac{x}{\sigma^2} \cdot e^{-\frac{x^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma^2} \cdot \int_0^{+\infty} x^2 \cdot e^{-\frac{x^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma^2} \cdot \frac{1}{2} \int_{-\infty}^{+\infty} x^2 \cdot e^{-\frac{x^2}{2\sigma^2}} dx$$

$(Y \sim N(0, \sigma^2))$ 的正态分布, $D(Y) = \sigma^2 = E(Y^2)$

而 $E(Y^2)$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} \cdot y^2 \cdot e^{-\frac{y^2}{2\sigma^2}} dy = \sigma^2$$

$$\therefore \int_{-\infty}^{+\infty} y^2 \cdot e^{-\frac{y^2}{2\sigma^2}} dy = \sqrt{2\pi} \sigma^3$$

$$\begin{aligned} \therefore E(X) &= \frac{1}{\sigma^2} \cdot \frac{1}{2} \times \sqrt{2\pi} \sigma^3 \\ &= \sqrt{\frac{\pi}{2}} \sigma \end{aligned}$$

$$D(X) = \underline{E(X^2)} - (\underline{E(X)})^2$$

$$\begin{aligned} \text{而 } E(X^2) &= \int_{-\infty}^{+\infty} x^2 f(x) dx \\ &= \int_0^{+\infty} x^2 \cdot \frac{x}{\sigma^2} \cdot e^{-\frac{x^2}{2\sigma^2}} dx \\ &= \int_0^{+\infty} x^2 \cdot \frac{1}{2\sigma^2} \cdot e^{-\frac{x^2}{2\sigma^2}} dx^2 \end{aligned}$$

$$\left(\underline{x dx = \frac{1}{2} dx^2} \right)$$

$$(\text{令 } x^2 = t)$$

$$= \int_0^{+\infty} \frac{1}{2\sigma^2} \cdot t \cdot e^{-\frac{t}{2\sigma^2}} dt \quad \Leftrightarrow$$

参数为 $2\sigma^2$ 的指数分布的期望

$$= 2\sigma^2$$

$$\therefore D(X) = 2\sigma^2 - \frac{\pi}{2}\sigma^2$$

15. 解: $P(|X - EX| \geq 7.5)$

$$\leq \frac{D(X)}{7.5^2}$$

$$= \frac{2.5}{7.5^2}$$

$$= \frac{2}{45}$$

20. 解:

$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy$$

$$= \int_0^2 \int_0^2 x \cdot \frac{1}{8}(x+y) dx dy$$

$$= 7/6.$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy$$

$$= \int_0^2 \int_0^2 y \cdot \frac{1}{8}(x+y) dx dy$$

$$= 7/6.$$

$$E(X^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 f(x, y) dx dy$$

$$= \int_0^2 \int_0^2 x^2 \cdot \frac{1}{8}(x+y) dx dy$$

$$= 5/3.$$

同理 $E(Y^2) = 5/3$.

$$\therefore D(X) = E(X^2) - (EX)^2 = 11/36$$

$$D(Y) = E(Y^2) - (EY)^2 = 11/36.$$

$$\text{cov}(X, Y) = \underbrace{E(XY)} - E(X) \cdot E(Y)$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy$$

$$= \int_0^2 \int_0^2 xy \cdot \frac{1}{8}(x+y) dx dy$$

$$= 4/3$$

$$\text{cov}(X, Y) = E(XY) - EX \cdot EY = -\frac{1}{36}.$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{DX} \cdot \sqrt{DY}} = -\frac{1}{11}.$$

$$\Sigma = \begin{pmatrix} D(X) & \text{COV}(X, Y) \\ \text{COV}(X, Y) & D(Y) \end{pmatrix}$$

$$= \begin{pmatrix} 11/36 & -1/36 \\ -1/36 & 11/36 \end{pmatrix}$$

24. 解:

$$\rho_{UV} = \frac{\text{cov}(U, V)}{\sqrt{D(U)} \cdot \sqrt{D(V)}}$$

$$\begin{aligned} D(U) &= D(aX + bY) \\ &= a^2 D(X) + b^2 D(Y) + 2ab \text{cov}(X, Y) \\ &= a^2 D(X) + b^2 D(Y) \\ &= (a^2 + b^2) \cdot \sigma^2 \end{aligned}$$

$$\begin{aligned} D(V) &= D(aX - bY) \\ &= a^2 D(X) + b^2 D(Y) - 2ab \text{cov}(X, Y) \\ &= a^2 D(X) + b^2 D(Y) \\ &= (a^2 + b^2) \cdot \sigma^2 \end{aligned}$$

$$\text{cov}(u, v)$$

$$= \text{cov}(\underline{aX} + bY, \underline{aX} - bY)$$

$$= a^2 \cdot \text{cov}(X, X) - ab \cdot \text{cov}(X, Y)$$

$$+ ba \cdot \text{cov}(Y, X) - b^2 \cdot \text{cov}(Y, Y)$$

$$= a^2 \cdot D(X) - 0 + 0 - b^2 \cdot D(Y)$$

$$= a^2 \cdot \sigma^2 - b^2 \cdot \sigma^2$$

$$= (a^2 - b^2) \cdot \sigma^2$$

$$\therefore \rho_{uv} = \frac{\text{cov}(u, v)}{\sqrt{D(u)} \cdot \sqrt{D(v)}} = \frac{a^2 - b^2}{a^2 + b^2}$$